Project Y

Los Alamos · 1943–45

15 STAGES · 55 STOPS · GRADE 12 · PROJECTY

THE OPENING CARD

It is April 1943, and the Army has built a secret lab on a mesa in New Mexico. You are a scientist at Project Y. Five teams work here, and each one uses its own words, its own tools and its own tests, so their numbers do not yet fit together. Over fifteen stages you sign The Evidence Chain: what the lab measured, what it worked out from that, and what nobody has checked yet. Get one link wrong and people far beyond the fence pay for it.

STAGE 1 OF 15

The Mesa Has No Common Language

BEFORE THE STAGE — GREET

The mesa is full of people who cannot tell you what they do

Meet the people in each division so you know who can help when the work begins.

PLAN CARD 110W

It is April 1943, and a secret lab now stands where a school once stood. Five teams use different words for the same things, so their results do not fit together. The director has asked for one shared language before anybody signs a number, and asked for it this week. The place is only weeks old, and it is already making calls it cannot take back. Nobody here has the time to redo a count they are not sure how to read. Today you must fix what each symbol and each signal means, and write it down. If you fail, later choices may rest on errors no one can see.

BRIEFING 46W

Los Alamos is only weeks old. Physicists, chemists, engineers, and military staff are using the same words to mean different things. Before the laboratory can compare evidence, the player must make the measurements commensurable: identify what was measured, in what units, and through which instrument chain.

WHAT YOU DECIDE 20W

Build a shared, checkable language for nuclei, isotopes, detector signals, and uncertainty before the new laboratory starts making irreversible decisions.

1.1 The nucleus as a physical system

THEORY & CALCULATIONS · A ROOM · PROTOCOL

SCENE 35W

Your first morning on the Hill. Oppenheimer, the laboratory director, stops at your bench with four nuclide labels and no explanation, and expects the right quantity pulled out of each one before anybody calculates anything.

GUIDE 52W

Four things you need and four ways to get them. Pair them by asking what each symbol counts. One counts protons. One counts protons and neutrons together. So one of these answers is a subtraction rather than a symbol on the label. And a neutral atom's electron count follows from its protons.

ASK 18W

Match each information need or situation to the best action, instrument, or control. Each option is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

an atom has a nucleus of protons and neutrons, with electrons around it

VERDICT 91W

Notation is only useful when each symbol is tied to a physical count. The atomic number Z is the proton count, and it fixes which element this is. The mass number A counts protons and neutrons together, so the neutron count is A minus Z — a subtraction, not a symbol on the label. Two atoms are isotopes of one element when Z is the same and A differs. And a neutral atom has as many electrons as protons, which is why Z decides the chemistry as well as the identity.

TAKEAWAY 11W

> Nuclear reasoning starts by translating notation into particles and conserved quantities.

1.2 Isotopes and chemical identity

CHEMISTRY & METALLURGY · A PERSON · PROTOCOL

SCENE 39W

Glenn Seaborg, the chemist charged with isolating plutonium, wants a straight answer from Chemistry and Metallurgy. It turns on one fact. Everything shipped here has a chemical form and an isotopic makeup, and they are not the same thing.

GUIDE 48W

Four questions and four answers, and three of them say yes. Ask of each whether the property lives in the electrons or in the nucleus. Bonding is electrons, and extra neutrons barely disturb them. Stability is nuclear. So ordinary chemistry has almost no handle here, and mass does.

ASK 18W

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TAKES AS READ

chemical bonding is done by electrons

VERDICT 84W

Isotopes of an element carry the same proton count and nearly the same electrons, so they form the same compounds and answer to the same reagents. Ordinary chemistry therefore has almost no handle for separating them. Nuclear stability is different: it is set inside the nucleus, and half-lives can differ enormously between isotopes of one element. Mass spectrometry sorts by mass-to-charge ratio, responding to the difference chemistry mostly ignores. That is why isotopic separation is a physical problem as well as a chemical one.

TAKEAWAY 10W

> Materials work must track both chemical form and isotopic composition.

1.3 Ionization and detector signals

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A ROOM · SEQUENCE

SCENE 41W

Robert Bacher, who heads the physics division, puts it plainly — nobody here can see a single neutron or alpha particle. Everything the Hill will claim about cross sections, backgrounds and yields rests on a chain of conversions working in order.

GUIDE 51W

These four are one chain, and a detector never measures the radiation itself. Ask of each card what has to exist before it. Nothing is amplified before something is set free. And nothing is set free before energy is deposited. That middle step decides what the instrument can see at all.

ASK 22W

Arrange the four cards from earliest cause or prerequisite to latest result, for a detector turning deposited energy into a recorded pulse.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

radiation deposits energy in whatever it passes through · decay constant, half-life and activity — taken as read

VERDICT 86W

A detector measures the consequences of an interaction, never the radiation itself. Energy has to be deposited in the sensitive material first — nothing downstream exists until that happens. That deposit sets free charge or light, which is the transducer step and the one that decides what the instrument is sensitive to at all. Electronics then collect and amplify a response too small to handle raw. The data system digitises and stores a pulse last. Get the chain wrong and you are measuring your own apparatus.

TAKEAWAY 11W

A detector measures the consequences of an interaction, not radiation directly.

END OF THE STAGE 16W

A large scientific program cannot integrate evidence until measurements share definitions, units, and traceable physical meaning.

Where Does the Energy Come From?

PLAN CARD 95W

It is May 1943, and the lab has more plans than people or time. A strong equation can still mislead if its inputs or tools are wrong. The theory division wants each number traced back to its source before it goes in a notebook. Two rival designs are both asking for staff, and only one can have them first. Today you must trace an energy claim back to its source and decide what proof each plan would need. Your choice will shape where the lab puts its staff and its tests for the next year.

BRIEFING 47W

The laboratory can calculate enormous nuclear energies, but the numbers are only useful if the instruments and assumptions behind them are sound. The player checks the mass-energy scale, breaks a calibration degeneracy, and compares the evidence demands of two historically competing design programs at a high level.

WHAT YOU DECIDE 19W

Connect mass, binding energy, calibration, and competing early program concepts without confusing a convincing calculation with a validated measurement.

2.1 Mass defect and binding energy

THEORY & CALCULATIONS · A ROOM · BALLPARK

SCENE 39W

Hans Bethe, who heads the Theoretical Division, wants a number before lunch: how tightly a medium-heavy nucleus is bound. The arithmetic is small enough for one line, but the energy scale is the reason this laboratory exists at all.

GUIDE 48W

Four numbers, and two of them are the same defect written a thousand times apart. Ask of each whether it is per nucleon or per nucleus. The conversion turns mass into energy, so it belongs beside the mass term. And work out the per-nucleon figure before the total.

ASK 11W

Estimate the total binding energy and the binding energy per nucleon.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

mass and energy are related by $E = mc^2$, with E the energy, m the mass and c the speed of light

VERDICT 88W

A bound nucleus weighs less than the loose protons and neutrons that went into it. That missing mass is the binding energy holding it together. The conversion factor is the speed of light squared. So a mass difference far too small to weigh becomes an energy larger than any chemical reaction can produce. Work in atomic mass units and 1 unit is about 931 MeV. That is what turns a defect of 8 thousandths of a unit per nucleon into more than 1000 MeV for the whole nucleus.

TAKEAWAY 9W

Tiny changes in mass correspond to large nuclear energies.

2.2 One reference line is not a calibration

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A PERSON · DEGENERACY

SCENE 41W

Two Physics groups bring spectra from different instruments to Bacher's afternoon comparison meeting, and the peaks refuse to line up. Each group trusts its own axis. Somebody has to decide whether the disagreement belongs to the samples or to the instruments.

GUIDE 49W

Two controls, gain and offset, slide along a family of calibrations that all put one reference line where it belongs. Find out how wide that family is before deciding anything. Then bring in the second reference line, which does not permit the same trade, and watch the family collapse.

ASK 10W

Which gain and offset are fixed by both reference lines?

BEHIND THE BUTTON 151W

Why one line cannot calibrate an axis. A linear axis has two unknowns: how much a step is worth, and where zero sits. One known peak gives one equation. Every gain paired with the offset that puts that peak in the right place fits equally well, which is exactly the family the controls slide along.

Why the second line is not a repeat. It constrains the same two unknowns through a different combination, so the pairs that satisfied the first no longer satisfy this one. Where the two constraints cross is the calibration, and it is why standards come with more than one line.

What is at stake this afternoon. If the axes can be reconciled, the spectra agree and the disagreement was instrumental. If they cannot, the difference is in the samples. Both groups trusting their own axis is how a meeting spends an afternoon arguing about the wrong thing.

TAKES AS READ

a straight-line calibration can have both a slope and an offset · one equation cannot determine two unknowns

VERDICT 70W

A straight-line calibration has two unknowns: gain and offset. One known line gives only one equation, so many gain-and-offset pairs reproduce it exactly. That is not noise; it is missing information. A second line at a different energy supplies a second equation. The two allowed loci cross at one point, fixing both parameters. This is why a convincing-looking one-point calibration can still place every unknown peak at the wrong energy.

TAKEAWAY 18W

One calibration point cannot determine both a slope and an offset; a second independent point breaks that degeneracy.

2.3 What each early program would have to prove

ORDNANCE & ENGINEERING · A ROOM · PROTOCOL

SCENE 40W

At a 1943 meeting, laboratory director Oppenheimer weighs two plans. One uses ideas the teams know well. The other needs new tests of shape and timing. You must decide what proof each plan needs before people and rooms are assigned.

GUIDE 49W

Four situations and four responses, and both designs chase the same goal. Ask of each whether it is about the geometry of the assembly or about its timing. One route drives two pieces along a line. One squeezes inward, and nobody has built one. A measurement decides between them.

ASK 22W

Match each high-level program claim to the kind of evidence it would require, without turning either historical architecture into a construction lesson.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

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TAKES AS READ

a chain reaction needs the material held together long enough to grow · pressure, temperature and equations of state — taken as read

VERDICT 77W

The important historical distinction is not a recipe for either device. It is that the two approaches imposed different evidence burdens on Los Alamos. The gun-type program relied on a comparatively mature linear-assembly concept. The implosion program required the laboratory to learn how to measure and control a three-dimensional compression process. That difference reorganized staffing, diagnostics, facilities and risk. Later plutonium measurements did not make the gun idea 'bad physics'; they made it unsuitable for that material.

TAKEAWAY 22W

Weapon architecture changed the laboratory's evidence burden, staffing and test program; the historical lesson is the program consequence, not the construction detail.

2.4 Radioactive is not the same as fissile

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A ROOM · BELT

SCENE 39W

A safety briefing has run three ideas together: radioactivity, neutron interaction and fissile behavior. Graves, a critical-assemblies physicist, wants the labels pulled apart. Somebody in the room has already read a loud counter as proof of a chain reaction.

GUIDE 50W

Two bins. A nuclide that fissions on slow neutrons will hold a chain given enough mass and the right geometry. One that only fissions on fast neutrons, or not at all, will not — however heavy the piece is. Sort on the nuclide, not on how dangerous the label looks.

ASK 11W

Which examples illustrate fissile behavior, and which illustrate different nuclear properties?

BEHIND THE BUTTON 77W

Why the distinction is the whole of storage policy. Two pieces that each hold no chain can hold one together, so the rule about what may sit next to what is written from this sort. Nothing about mass or radioactivity substitutes for it.

Why radioactive and fissile are different words. Cobalt-60 will make a counter scream across the room and will never sustain anything. Natural uranium is barely active and contains the isotope everything here is about.

TAKES AS READ

some heavy nuclei split when they absorb a neutron · fission, neutron multiplication and chain reactions — taken as read

VERDICT 55W

Radioactivity means a nucleus changes on its own. Fissile means an absorbed neutron can help a chain of fissions grow in the right setting. These are different traits. Uranium-235 and plutonium-239 are fissile; the other examples show other nuclear or material traits. A strong counter reading alone does not show that a sample is fissile.

TAKEAWAY 18W

Radioactivity, neutron interaction and fissile behavior are different physical claims and must not be inferred from one another.

END OF THE STAGE 23W

A correct equation is only one link in an evidence chain; calibration and assumptions decide whether its result can support a program decision.

The Counts Do Not Agree

PLAN CARD 102W

It is July 1943, and three reports give three amounts for the same material. The sample changes with time, while the counter may add signals of its own. The physics division will not pick between the three reports until they share a clock. Each one was taken on a different day, at a different bench, with a different counter. No two of them were read by the same pair of eyes. Today you must put every result on one clock and find where the gap begins. Until you do, the lab may blame the wrong team and lose stock it cannot replace.

BRIEFING 38W

Three groups report incompatible yields from the same material stream. The player follows a decaying tracer through a separation, corrects measurements to a common time, and isolates a background-count problem before the disagreement gets mistaken for new physics.

WHAT YOU DECIDE 18W

Use decay, tracer recovery, and background measurements to determine whether conflicting laboratory yields are physics, chemistry, or instrumentation.

3.1 Radioactive decay law

THEORY & CALCULATIONS · A ROOM · BALLPARK

SCENE 38W

A tracer sample has sat through a six-hour chemistry shift, and the separation team needs to know what remains before the next measurement starts. Every count here belongs to something that has been decaying since it was made.

GUIDE 50W

Four numbers, and two of them are counts of half-lives that differ by a factor of ten. Ask of each whether it is a population, a fraction, or a number of halvings. Six hours of two-hour halvings is not thirty of them. And each halving multiplies by the same fraction.

ASK 20W

Estimate how many nuclei remain after the shift, then use the half-life to estimate the sample's activity at that moment.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a half-life is the time for half of a large population to decay · radioactive decay modes: alpha, beta, gamma — taken as read · exponential decay and logarithms — taken as read

VERDICT 73W

Decay is random for one nucleus and predictable for a large population. 6 hours is 3 two-hour half-lives, so 10,000 nuclei fall to 1 eighth, or about 1,250. Half-life also fixes the decay constant: $\lambda = \ln 2/t_{½}$. For a two-hour half-life, λ is about 9.6×10^{-5} s^{-1}. Activity is then $A = \lambda N$, so the remaining sample produces about 0.12 decays each second. Population and count rate are linked by the same decay law.

TAKEAWAY 19W

Half-life fixes both how a population falls with time and, through the decay constant, its activity at any moment.

3.2 Radiochemical tracers

CHEMISTRY & METALLURGY · A PERSON · BALLPARK

SCENE 31W

Chemistry & Metallurgy is running separations on quantities too small to weigh directly and needs to know how much is being lost between steps. A decaying tracer provides the bookkeeping signal.

GUIDE 50W

Four numbers, in two pairs, and only one pair belongs to this separation. Ask of each whether it is what went in or what came out. And note what makes a raw ratio dishonest here. The tracer decayed throughout, so both counts have to be corrected to the same moment.

ASK 4W

Estimate the chemical yield.

BEHIND THE BUTTON 106W

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TAKES AS READ

a radioactive sample announces itself through its own radiation

VERDICT 81W

A tracer is a small amount of radioactive material that travels with the chemistry and reports where it went. That turns an invisible loss into a measurable fraction: yield is what came out over what went in. The correction that makes it honest is decay — the tracer has been decaying throughout the separation, so a raw count afterwards understates the recovery and flatters the loss. Both counts have to be corrected to the same moment before the ratio means anything.

TAKEAWAY 10W

A tracer turns invisible material loss into a measurable fraction.

3.3 Find what carries the background

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A ROOM · CONTROL

SCENE 26W

A weak sample has been counting for an hour, and Elizabeth Graves wants to know whether the excess belongs to the sample or the counting hardware.

GUIDE 75W

The number you are watching is the counting rate with no sample in the station, and the excess is how much of it cannot be accounted for. A change has told you something only when it moves that excess by more than the rate wanders on its own between runs. Several parts of the station changed when it was built, so choose one, run it, and put it back. Name the part the excess follows.

ASK 9W

Which part of the counting station carries the excess?

BEHIND THE BUTTON 178W

What a background is. A detector counts things whether or not you gave it anything to count: cosmic rays, activity in the concrete, contamination on the housing, noise in the electronics. The rate with no sample in place is the sum of all of those, and a weak sample is only detectable to the extent that its own counts stand above that sum.

Why one change at a time. Change two things and a response tells you only that one of them mattered. The station has a shield, a room and a detector, and each of them can carry background for its own reason, so the only reading that names a part is the one where nothing else moved.

Why putting it back matters. A rate that drops when you shield the detector and stays down when you unshield it was falling for some other reason — drift, a decaying source, somebody switching off a pump down the corridor. A cause has to follow the change in both directions, and that is the whole of what reversal tests.

TAKES AS READ

background is measured with no sample present · one changed variable can be tested against a stable baseline

VERDICT 78W

The sample cannot explain a count excess that remains after the sample is removed. Extra shielding cannot explain it when more lead changes nothing. The decisive comparison is hardware: the original detector still runs high in a clean room, while an identical clean detector in the same room returns to the normal background. Put the original detector back and the excess returns. That reversal makes contamination of the detector or holder a causal result rather than a correlation.

TAKEAWAY 19W

A suspected cause is strongest when changing only that cause moves the reading and reversing the change restores it.

END OF THE STAGE 19W

Time-dependent samples and instrument backgrounds must be separated before a count rate can be treated as a material yield.

Is the Difference Real?

BEFORE THE STAGE — FOLLOW

Keep up with the commanding general

Keep up with the commanding general so you can see which parts of the lab need help.

PLAN CARD 96W

It is October 1943, and two teams disagree about a gap in their counts. The gap may be real, or it may be normal noise, or metal that went missing. The bench that runs the neutron counts has watched them drift for both of those reasons. The sample they are arguing over is rare, and there is no second one on the mesa. Today you must decide how much the gap proves, and make the metal ledger close. A bad call could waste that sample and send the lab after a result that is not there.

BRIEFING 35W

Two groups believe their measurements disagree. The player converts interaction probability into a measurable scale, asks whether the count difference exceeds ordinary statistical fluctuation, and refuses a chemistry result whose mass balance does not close.

WHAT YOU DECIDE 20W

Use cross sections, counting statistics, and material balance to decide whether an apparent disagreement is large enough—and clean enough—to matter.

4.1 Reaction cross sections

THEORY & CALCULATIONS · A ROOM · BALLPARK

SCENE 33W

Bethe needs a reaction rate before the afternoon seminar and hands you a cross section written in barns. Nearly every number this laboratory argues about is built out of the multiplication that follows.

GUIDE 53W

Four numbers, and two of them are areas thirty powers of ten apart. Ask of each whether it is a density, an area, or a bare one. Check the units before choosing, because a cross section is an area. One parameter comes out of the first step and then does all three jobs.

ASK 22W

Estimate the macroscopic cross section, the mean free path, and the fraction of an uncollided beam left after one mean free path.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a cross section is expressed as an area · fission, neutron multiplication and chain reactions — taken as read

VERDICT 76W

A microscopic cross section is an interaction probability written as an area. Multiply it by the number of targets per volume and you get the macroscopic cross section Σ, an inverse length. Its inverse is the mean free path. The same Σ also controls attenuation: $I/I_0 = e^{-\Sigma x}$. After 1 mean free path, $x = 1/\Sigma$, so the exponent is minus 1 and about 37% of an uncollided beam remains. One parameter connects all three ideas.

TAKEAWAY 22W

Number density turns a microscopic cross section into an interaction probability per length, which sets both mean free path and exponential attenuation.

4.2 Poisson counting statistics

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A PERSON · BALLPARK

SCENE 38W

Joan Hinton, the young experimenter running the counter, has finished a run with 400 net counts and wants to know how well the result is known. Beam time, detector time, material, and shift hours are all scarce here.

GUIDE 47W

Four numbers, and three of them are count totals. Only one belongs to this run. Ask of each whether it is the numerator or the total. And note how slowly this improves. Halving the uncertainty needs four times the counts, which makes run length a planning decision.

ASK 5W

Estimate the fractional statistical uncertainty.

BEHIND THE BUTTON 106W

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TAKES AS READ

radioactive decays arrive at random · descriptive statistics: mean, spread, standard deviation — taken as read

VERDICT 72W

A repeat of the same run would not give exactly the same total. For Poisson counting, the typical fluctuation is about the square root of the count, so the fractional uncertainty is $1/\sqrt{N}$. 400 counts gives 5%. To cut that in half, to 2.5%, you need 4 times as many counts. That slow square-root improvement is why counting statistics are a planning tool: longer runs buy precision, but they buy it expensively.

TAKEAWAY 13W

Four times as many counts are needed to cut statistical uncertainty in half.

4.3 Close the material balance

CHEMISTRY & METALLURGY · A ROOM · BALANCE

SCENE 35W

A separation batch has come back and the numbers do not close. Chemistry & Metallurgy will not release the report until the ledger does; downstream work assumes the reported recovery really means what it says.

GUIDE 49W

Read every stream first, because reading costs nothing. Then count only the masses that belong to the same material ledger. The batch does not close, and the missing mass has gone somewhere real — an uncounted route rather than an arithmetic slip. Report the total when the ledger closes.

ASK 7W

Which uncounted route closes the material balance?

BEHIND THE BUTTON 130W

Why a balance has to close. Material is conserved, so a batch that does not add up is a statement that something has not been measured, not that some of it stopped existing. Every unclosed ledger is a location nobody has looked at yet.

Where material actually goes. Holdup in the equipment, residue on filters and vessel walls, a side stream taken for analysis, losses to a scrubber. Each is ordinary and each is invisible in the streams somebody thought to weigh.

Why it matters downstream. Somebody is going to cast this material assuming the report means what they think it means. In this programme, an assay that is quietly wrong about how much is present is the kind of error that ends with a criticality rather than a memo.

TAKES AS READ

percent of input material can be added across mutually exclusive output routes · purity and recovered quantity are different measurements

VERDICT 69W

The input is 100% of the target material. Recorded product and aqueous output account for only 93.9%, so 6.1% is still missing. The 91% purity result answers a different question and cannot be added to that ledger. A wall swab finds 5.8% of the input on the vessel surface. Add that loss route and the balance closes to 99.7%, inside rounding. Yield, purity, and material balance are different claims.

TAKEAWAY 19W

A purity result cannot substitute for a closed material balance; losses outside measured process streams must still be counted.

END OF THE STAGE 19W

A scientific disagreement becomes actionable only after random fluctuation, missing material, and shared measurement errors have been ruled out.

The Material Changes the Plan

PLAN CARD 105W

It is spring 1944, and new plutonium gives far more stray neutron counts than expected. The counts could come from the metal itself, or from counters that have gone bad. The chemists on the plutonium work say the new samples are not like last year's. Both groups are sure of their reading, and they cannot both be right. The lab has spent a year on a plan that assumed the metal would not change. Today you must use independent evidence to decide which of the two readings is true. If the metal has changed, that year of work may no longer solve the real problem.

BRIEFING 43W

New material from the production reactors does not behave like the earlier samples. The player separates neutron-transport physics from detector failure and then interprets the historically decisive result: reactor-produced plutonium carried a much higher spontaneous-neutron background than the plutonium gun program could tolerate.

WHAT YOU DECIDE 21W

Use neutron transport and independent detector evidence to understand why reactor-produced plutonium forced Los Alamos to reconsider a major design path.

5.1 Neutron energy and moderation

THEORY & CALCULATIONS · A ROOM · PROTOCOL

SCENE 39W

Bethe sketches a fission neutron on the board and asks what happens next — it carries a great deal of energy, and everything depends on what it hits. Confusing the possibilities is how a shielding calculation goes quietly wrong.

GUIDE 49W

Four things that can happen to a neutron, and four processes. Pair them by asking two questions. Is the neutron still there afterwards? And how much energy did it give up? A heavy target barely recoils, so it takes almost nothing away. One of these removes the neutron entirely.

ASK 18W

Match each information need or situation to the best action, instrument, or control. Each option is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a light ball loses more energy hitting something its own size · fission, neutron multiplication and chain reactions — taken as read

VERDICT 87W

Slowing, redirecting and absorbing are three different processes with three different consequences. In elastic scattering the neutron transfers some energy and continues as a neutron — it is still in the system. In absorption it is captured into a compound nucleus and is gone. Scattering off a heavy nucleus changes direction and gives up very little energy, because a heavy target barely recoils. Scattering off light nuclei is what slows neutrons efficiently, because a light nucleus can take a large share of the energy in one collision.

TAKEAWAY 8W

Energy management and neutron removal are not interchangeable.

5.2 Are the neutrons missing, or the counts?

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A ROOM · TRACE

SCENE 41W

A neutron counter is reading well below prediction, and Bacher wants to know whether the physics or the instrument is at fault before anybody reports it upstairs. The measurement built on this counter is one the whole plutonium programme depends on.

GUIDE 51W

Open each reading to see what it was computed from. Keep the ones that could still see neutrons arriving without passing through the discriminator, and untick the rest. Then name the dependency the failed readouts share. Both halves count: discarding an independent measurement costs as much as keeping a dependent one.

ASK 12W

Are neutrons failing to arrive, or is the counting chain discarding them?

BEHIND THE BUTTON 127W

Why the two explanations look identical on the console. Fewer neutrons arriving and fewer neutrons being counted produce the same low number. Nothing about the number itself separates them, which is why the answer has to come from what each channel depended on rather than from its value.

What the counting chain contains. A detector, a discriminator that decides whether a pulse is large enough to count, an amplifier, a scaler. A threshold set too high silently rejects real events, and every channel downstream of it reports the same shortfall convincingly.

Why an independent channel is worth so much here. An activation foil or a geometry check sees neutrons arriving without the electronics deciding anything. One such measurement settles a question no number of counter runs can.

TAKES AS READ

a neutron detector converts an interaction into an electrical pulse · a discriminator rejects pulses below a set threshold

VERDICT 79W

The main counter is low, but that result depends on every stage of the counting chain. The activation foil does not: it sees the expected neutron exposure without using the discriminator electronics. Geometry also behaves as expected under the inverse-square check. Meanwhile the pulse-height distribution piles up just below the discriminator threshold. Those observations can all be true together only if neutrons are arriving and real detector pulses are being rejected. Agreement among dependent readouts is not independent evidence.

TAKEAWAY 17W

Independent evidence can separate a missing physical signal from a failure in the instrument that reports it.

5.3 Why plutonium required implosion

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A PERSON · PROTOCOL

SCENE 41W

Measurements on reactor-produced plutonium have come back from the counters, and Oppenheimer has called Ordnance in. The result changes not one component but the direction of the entire programme, and the reason has to be understood before the reorganization makes sense.

GUIDE 50W

Four situations and four responses. Ask of each whether it is about the material or about the engineering. This material emits neutrons on its own, at a rate nobody chose. So the clock every assembly has to beat is set by what comes out unprompted, not by a design decision.

ASK 16W

Match each situation to the best action, control, document, or interpretation. Each option is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a chain reaction started early releases energy early · pressure, temperature and equations of state — taken as read

VERDICT 88W

Reactor-produced plutonium contains isotopes that emit neutrons on their own. The material supplies its own stray neutrons, at a rate nobody chose. A slow assembly spends longer in the state where one of those could start the chain. A chain that starts early releases enough energy to blow the pieces apart before most of the material has reacted. That is pre-initiation. It is a property of the material, not a

> defect better workmanship can fix. Implosion is the answer: squeeze from all sides, and cut the vulnerable time.
>
> **TAKEAWAY** 8W
>
> Material physics can determine the required system architecture.

5.4 Isotopes and chemical identity — Review

CHEMISTRY & METALLURGY · A ROOM · CALLBACK · PROTOCOL

SCENE 42W

Lilli Hornig, the chemist working on lenses and plutonium, has four questions from the stockroom on one sheet. Each turns on the same thing. Does the property being asked about belong to the electrons of the delivered material, or to its nucleus?

GUIDE 53W

Four questions and four places to look. Ask of each whether the property lives in the electron shells or in the nucleus. Bonding and dissolving are electrons, and a few extra neutrons barely disturb them. What an arriving neutron does is nuclear. And a physical separation has only the mass to work with.

ASK 18W

Match each information need or situation to the best action, instrument, or control. Each option is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

chemical bonding is a property of the electrons

VERDICT 81W

Chemical behaviour is decided by the electron shells, and the electron count follows from the proton count alone. Two isotopes therefore carry the same shells, dissolve in the same acid, and behave alike in every reaction, which is why no chemical assay separates them. What an arriving neutron does is decided in the nucleus, and there the two are not alike at all. The one handle left is mass, and every separation plant built for this project works on nothing else.

TAKEAWAY 15W

> Which method can see a difference depends on where in the atom that difference lives.

END OF THE STAGE 25W

When a material property invalidates a design assumption, the responsible response is to verify the property independently and then change the program—not defend sunk work.

STAGE 6 OF 15

A Year of Work Ends

PLAN CARD 96W

It is July 1944, and several tests now show that the old plutonium plan rests on a false claim. Ending it will throw away a year of work and move people to a less tested plan. The director will not close a whole program on the strength of one result. Groups will be broken up, and the people who led them have to be told why. Today you must show what failed and decide how the lab should change course. The new plan must answer to evidence, not to pride or to a year already spent.

BRIEFING 42W

The higher spontaneous-neutron background is confirmed. The player estimates a laboratory background count, follows the evidence chain that ended the plutonium gun program, and sequences the reorganization needed to test a different approach without treating the device itself as an instructional object.

WHAT YOU DECIDE 23W

Turn the 1944 plutonium result into a program decision while keeping the lesson at the level of evidence, timing risk, and organizational change.

6.1 How large is the neutron background?

THEORY & CALCULATIONS · A ROOM · BALLPARK

SCENE 35W

The new reactor-produced plutonium gives a neutron background far above the earlier material. Before anyone connects that fact to a program decision, Bacher wants the size of the background written as an ordinary counting problem.

GUIDE 44W

Rate times time gives an expected count. This is a laboratory-background calculation, not a device-timing calculation. Pick the measured rate and the observation time, then check whether the result is large enough that ordinary Poisson fluctuation cannot explain the difference from the old samples.

ASK 27W

Use the measured excess count rate and the counting interval to estimate the excess counts, then weigh them against the statistical uncertainty on a count that size.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

some heavy nuclei split without being hit by anything · descriptive statistics: mean, spread, standard deviation — taken as read

VERDICT 79W

A count rate becomes comparable only when it is tied to an observation time. Here the new sample averages 120 counts per minute above background for 10 minutes, so about 1,200 excess counts are expected. The square-root statistical fluctuation of a count this large is only a few tens of counts, far smaller than the excess. That does not by itself choose a weapon architecture; it establishes that the material property is real enough to force the next question.

TAKEAWAY 13W

The decisive historical fact was a reproducibly higher spontaneous-neutron background in reactor-produced plutonium.

6.2 Why the plutonium gun program ended

ORDNANCE & ENGINEERING · A PERSON · SEQUENCE

SCENE 25W

The higher spontaneous-neutron background has now been reproduced. Oppenheimer must decide what it means for the plutonium gun program and how to reorganize the laboratory.

GUIDE 34W

Keep this at the level of assumptions and program consequences. The new material measurement comes first. Then identify which earlier assumption it invalidates, make the program decision, and only then reorganize staff and tests.

ASK 9W

Order the historical evidence-to-decision chain without reconstructing the device.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a measurement can rule out a design without anybody being wrong · pressure, temperature and equations of state — taken as read

VERDICT 55W

Reactor-produced plutonium carried a much higher spontaneous-neutron background than the early program had assumed. That made the plutonium gun approach unreliable. Los Alamos therefore abandoned that route for plutonium and concentrated the plutonium effort on implosion research. The historical lesson is how a new measurement can invalidate a design assumption and force a program-level pivot.

TAKEAWAY 23W

The plutonium gun program ended because a newly measured material property invalidated a key assumption, not because the underlying nuclear physics suddenly changed.

6.3 How a laboratory survives a design pivot

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A ROOM · SEQUENCE

SCENE 25W

Oppenheimer has ended one major plutonium line of work. People, facilities and diagnostics now have to move without losing the evidence that caused the change.

GUIDE 36W

This is a program-recovery sequence, not a device sequence. Preserve the failed assumption and its evidence, define what the replacement approach must demonstrate, build measurements that could falsify it, and only then commit the integrated schedule.

ASK 14W

Arrange the program steps that turn a failed assumption into a testable replacement program.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a detonation travels outward from where it starts · pressure, temperature and equations of state — taken as read

VERDICT 57W

A pivot is more than moving people. The laboratory first records the failed assumption and the measurements that disproved it. It then states the new program-level requirements without pretending the replacement is already validated. Independent diagnostics are built around those requirements. Only after the tests can actually distinguish success from failure does the integrated schedule become meaningful.

TAKEAWAY 22W

A successful pivot preserves the evidence that killed the old plan and gives the new plan explicit tests it can still fail.

6.4 Ionization and detector signals — Review

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A ROOM · CALLBACK · SEQUENCE

SCENE 43W

Leona Woods, who builds the counters here, has a tape of raw counts from an overnight run and a figure somebody has already quoted from it. She wants the steps between the two written out in order before that figure goes any further.

GUIDE 61W

These four are one chain, and the number at the end is meant to belong to the source rather than to the counter. Ask of each card what has to exist before it. Nothing can be corrected as a rate until it is a rate. And nothing about the source can be claimed while the instrument is still inside the number.

ASK 12W

Arrange the four cards from earliest cause or prerequisite to latest result.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a counter records only a fraction of the events that happen · decay constant, half-life and activity — taken as read

VERDICT 85W

A counter reports its own events, never the source. The pulses are totalled first, then divided by the live time, because every correction after that is written as a rate. Next the background rate comes out, measured with the source taken away, since the counter answers to cosmic rays and to contamination whether or not anything is on the shelf. Only then does efficiency convert what was detected into what actually decayed. Skip a step and the figure quoted is a property of the apparatus.

TAKEAWAY 19W

A reported quantity belongs to the source only once everything belonging to the instrument has been taken back out.

END OF THE STAGE 25W

A scientific pivot is strongest when the failed assumption is named explicitly and the new program is organized around tests that can falsify its replacement.

Can Compression Be Predicted?

PLAN CARD 101W

It is September 1944, and the new plan leans on matter at pressures no table covers. A metal shifts form, and two tools disagree about an inert test. The explosives division wants a test it can repeat before it will trust a model. Nobody has yet run the same shot twice and got the same trace. The danger is not what the lab does not know, but what it thinks it knows. Today you must decide what the models truly support, and which new check is needed. Guess here and every later shot rests on a number that nobody ever measured.

BRIEFING 47W

The new program depends on matter behaving predictably under extreme compression, but equations of state and phase behavior are uncertain outside ordinary laboratory conditions. The player checks compression work, diagnoses a phase-model mismatch, and designs an inert wave-shaping measurement that can test symmetry without revealing weapon-construction details.

WHAT YOU DECIDE 15W

Connect high-pressure material behavior, phase changes, and inert wave-shaping experiments while keeping model limits visible.

7.1 Equations of state

THEORY & CALCULATIONS · A ROOM · BALLPARK

SCENE 45W

Von Neumann, a mathematician on the implosion work, is calculating work done on a material at pressures nothing in the laboratory has ever measured, and wants the units checked line by line. An error in the units looks exactly like an error in the physics.

GUIDE 46W

Four numbers: two pressures and two volumes, each pair a hundredfold apart. Ask of each whether it describes this regime or handbook conditions. Work is a pressure times a volume change, so an error in either is an error of the same size in the answer.

ASK 4W

Estimate the work scale.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

pressure times a volume change is work · units and conversions: MeV, u, barns, scientific notation — taken as read

VERDICT 88W

An equation of state ties pressure, density, temperature and internal energy together for a substance. The familiar handbook values were all taken at room conditions. They describe a completely different regime from the one implosion makes. That is why this work is unforgiving in two directions at once. The thermodynamics has to be right. So does every unit conversion carrying it across many powers of ten. Pressure in pascals times a volume change in cubic metres gives joules, and the powers of ten do most of the work.

TAKEAWAY 10W

Extreme-state calculations require both thermodynamic relationships and careful unit scaling.

7.2 Phase diagrams

CHEMISTRY & METALLURGY · A PERSON · DIAGNOSIS

SCENE 41W

Hornig, a chemist studying plutonium and test materials, has a microstructure under the microscope that the phase diagram says should not be there. Everything the laboratory does with this metal is an argument with a diagram drawn from very little data.

GUIDE 53W

A phase diagram is drawn for one composition. Ask of each candidate whether it questions the map, or what the map was drawn for. Plutonium has several solid phases and large volume changes between them, so a crossed boundary is a real effect. But check you still have the material the diagram describes.

ASK 19W

Read the whole reading panel, including the calm readings, then select the one cause that fits all of it.

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a metal can exist in more than one solid form · units and conversions: MeV, u, barns, scientific notation — taken as read

VERDICT 85W

A phase diagram maps which solid and liquid phases are stable at each temperature and composition. Crossing a boundary can melt a material, throw out a second phase, or change how it behaves under load. Plutonium is notorious for it. Several solid phases, and large volume changes between them. But a diagram describes one composition. So before doubting the map, check you still have the material it was drawn for. A trace impurity is enough to hold open a phase that should not be there.

TAKEAWAY 9W

Composition and temperature jointly determine which phases are possible.

7.3 What an inert wave-shaping test can establish

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A ROOM · PROTOCOL

SCENE 38W

Kistiakowsky, the head of the explosives division, has his group studying how waves move through different materials. The player is not designing the weapon; the task is to decide which measurements make a claim about wave shape defensible.

GUIDE 29W

Pair each claim with the observation that isolates it: measure material propagation separately, then combined arrival times, then manufacturing repeatability, then cross-check the reconstructed front with an independent diagnostic.

ASK 16W

Match each wave-shaping question to a measurement that can answer it in an inert hydrodynamic mockup.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a lens bends light by slowing it in some places more than others · pressure, temperature and equations of state — taken as read

VERDICT 52W

Pressure waves move at different speeds in different materials. An inert test can show how even their arrival is, how much joints or batches change it, and whether a second tool sees the same front. Each claim needs its own measure. The test supports only the parts that those measures can check.

TAKEAWAY 16W

Wave-shaping claims are earned by inert measurements of propagation, arrival-time uniformity, manufacturing variation, and independent reconstruction.

END OF THE STAGE 16W

Extreme-condition models need independent material measurements and inert validation tests before their extrapolations can be trusted.

Make 'Nearly Simultaneous' Measurable

BEFORE THE STAGE — HUNT

Five detectors signed out, and the log accounts for three

Find the two missing detectors so every later count can be tied to the right instrument.

PLAN CARD 103W

It is October 1944, and three teams use the words "nearly at once" in three different ways. A vague claim cannot guide a test or show that a system works. Nobody can yet tell the theory group how near is near enough, or why. No one has put a number on it, or on how well that number is known. A test cannot be built on a phrase that three teams read three ways. Today you must choose a shared measure, show how to read it, and state its doubt. After this, every timing claim must come with a number and a check.

BRIEFING 44W

Different groups are using words such as 'simultaneous' and 'symmetric' without a shared quantitative test. The player dates a decaying source, traces timing jitter through an instrument chain, calculates a cable-delay mismatch, and controls a numerical chain-reaction model rather than operating real critical hardware.

WHAT YOU DECIDE 15W

Turn vague claims about timing and symmetry into calibrated measurement requirements and explicit uncertainty budgets.

8.1 How many half-lives have passed

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A ROOM · BALLPARK

SCENE 36W

A sample counted at 8,000 counts a minute reads 500 today, and the notebook does not say when the first count was taken. Kistiakowsky wants the elapsed time; the chart puts the half-life at 12 hours.

GUIDE 52W

Six numbers, and three of them are steps of the working: a ratio, a count of halvings, and a base. Ask of each whether the relationship needs it or produces it. And note the habit. Work in ratios rather than differences, because a drop of 7,500 counts means nothing on its own.

ASK 8W

Estimate the elapsed time since the first count.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a half-life is the time for half of a sample to decay

VERDICT 81W

Every half-life halves the count rate, so the elapsed time is the number of halvings times the half-life. 8000 to 500 is a factor of 32 is 2 to the fourth: 4 half-lives, or 48 hours. Written as a logarithm it is log base two of the ratio, which is what a semi-log plot shows as a straight line. The useful habit is to work in ratios rather than differences — the difference of 7,500 counts means nothing on its own.

TAKEAWAY 16W

Exponential decay is counted in half-lives, and logarithms are how you get the number of them.

8.2 Find where timing jitter enters

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A ROOM · CONTROL

SCENE 34W

A timing array is disagreeing with itself, and Bacher has ruled that until this is settled no symmetry claim from it means anything. The Hill is about to spend months measuring differences of nanoseconds.

GUIDE 69W

The number you are watching is the spread in arrival times between channels, and the live path gives about fourteen nanoseconds of it. Substituting a reference part asks a specific question: if that part were perfect, how much spread would be left? A substitution that leaves the spread where it was has cleared that part. Run one at a time, restore it, and name the stage the spread follows.

ASK 14W

Which part of the live timing path is responsible for the broad arrival-time spread?

BEHIND THE BUTTON 165W

What the spread is made of. Every stage between the particle and the recorded time adds its own variation: how the detector forms a pulse, how the cable carries it, how the discriminator decides when the pulse crossed a threshold. Those add in quadrature, so the largest one dominates and the others are nearly invisible until it is removed.

Why substitution rather than adjustment. A reference pulse injected at the electronics input is a known, jitter-free signal. Feeding it in replaces one stage with something whose contribution is negligible, so what remains is everything else. That makes the measurement subtractive: the drop when a stage is bypassed is that stage's own contribution.

What is riding on it. Months of measurement here are differences of nanoseconds, so a timing path that adds fourteen of them is not a detail. Until the array agrees with itself, no asymmetry it reports can be separated from its own jitter, which is why this is settled before the physics starts.

TAKES AS READ

a fixed delay shifts a distribution while random jitter broadens it · a controlled substitution changes one part of a measurement chain at a time

VERDICT 81W

A useful control changes one suspected part of the measurement chain while leaving the rest in place. The live timing path shows a 14 ns spread. Equalized cables or a replacement discriminator leave nearly the same broad distribution, so neither explains it. Identical pulses injected at the electronics inputs collapse the spread to about 1 ns. When the live detector path is restored, the broad spread returns. The changing element is therefore the live pulse-formation stage upstream of the injection point.

TAKEAWAY 21W

A controlled substitution localizes an instrument error only when one change removes the effect and reversing that change brings it back.

8.3 Why synchronization became an engineering requirement

ORDNANCE & ENGINEERING · A PERSON · BALLPARK

SCENE 38W

Two fast diagnostic channels are supposed to observe the same inert event, but one signal path is longer. Parsons, an ordnance engineer, wants the delay budget written in a form a technician can inspect with a tape measure.

GUIDE 24W

A path difference becomes an arrival-time difference because signals travel at a finite speed. Choose the path mismatch, signal speed, and conversion to nanoseconds.

ASK 9W

Estimate the timing difference created by unequal diagnostic-cable lengths.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a signal travels along a cable at a finite speed

VERDICT 55W

A small path-length difference in a fast diagnostic system becomes a measurable timing offset. Divide the extra cable length by the signal speed. This turns a vague requirement such as 'record together' into something inspectable. The lesson is metrology: a timing claim is only as good as the physical paths and references that define it.

TAKEAWAY 16W

Synchronization is a physical measurement-chain problem: cable length, electronics, and reference clocks all contribute timing error.

8.4 Hold a multiplication model at steady state

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A ROOM · HOLD

SCENE 34W

Bethe has put a simplified multiplication model on the board. The purpose is to understand feedback, not to operate a real reactor. A slider changes the model's loss term while several abstract parameters drift.

GUIDE 34W

Keep the simulated level inside the band by balancing modeled growth and loss. Treat every disturbance as a change in a rate. The exercise teaches feedback and dynamic equilibrium without using real reactor controls.

ASK 13W

Hold the simulated multiplication level near its target while the model coefficients drift.

BEHIND THE BUTTON 84W

In a multiplication model, a level grows when each generation produces more than replaces itself and falls when losses dominate. A steady trace therefore represents a balance of rates, not a system that has stopped changing.

The disturbances here are deliberately abstract model coefficients. They stand for any slow parameter drift in a feedback system and are not instructions for controlling nuclear hardware.

The useful habit is to anticipate a persistent rate change rather than chase the displayed level after it has already moved.

TAKES AS READ

a multiplication model can grow, hold steady, or decay depending on the balance of gain and loss · radioactive decay modes: alpha, beta, gamma — taken as read

VERDICT 52W

The exercise is a numerical feedback model. If modeled gain exceeds modeled loss, the displayed level grows; if loss exceeds gain, it decays. A steady level therefore requires the two rates to balance. The control is intentionally abstract so the lesson is about dynamic equilibrium and feedback, not about operating a reactor.

TAKEAWAY 25W

A dynamic steady state is a balance of rates, and feedback should be reasoned about through the model rather than by copying real hardware procedures.

END OF THE STAGE 16W

Engineering language becomes scientific evidence only when qualitative requirements are tied to calibrated observables and uncertainty.

STAGE 9 OF 15

Prove It Without the Real Device

PLAN CARD 100W

It is December 1944, and the lab cannot test the final system again and again. Each stand-in can answer one question but may hide what it cannot copy. The field-test group has one mockup on the bench and a long list of claims to check. Every stand-in test also spends metal the mesa cannot get more of. A mockup trusted too far is worse than no mockup at all. Today you must set material limits, judge a mockup, and measure motion during the test. If you claim too much, the final plan may rest on proof the stand-in never gave.

BRIEFING 44W

A full nuclear system cannot be used as an iterative laboratory test. The player therefore has to decide what individual material tests and inert mockups are entitled to prove, and where conservative acceptance criteria are needed because the real system cannot be repeatedly tested.

WHAT YOU DECIDE 20W

Use material tests, conservative acceptance rules, inert mockups, and fast diagnostics to learn from stand-ins without pretending they reproduce everything.

9.1 Mechanical properties

CHEMISTRY & METALLURGY · A ROOM · DIAGNOSIS

SCENE 40W

A component has failed in service and Hornig has put the failure analysis on your bench. This programme cannot spare either the material or the weeks, so the question is what changed between the qualified coupon and this actual part.

GUIDE 55W

Four properties, and none of them is a synonym for another. Ask of each what it actually measures. Deflection under load? Where permanent bending begins? Resistance to being dented? Or energy absorbed before it breaks? A material can be hard and brittle, and that is how a part passes a test and fails in service.

ASK 19W

Read the whole reading panel, including the calm readings, then select the one cause that fits all of it.

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a material can be strong in one sense and weak in another · units and conversions: MeV, u, barns, scientific notation — taken as read

VERDICT 85W

Four properties are involved and they are not synonyms. Elastic modulus is stiffness — how much it deflects under load. Yield strength is where permanent deformation begins. Hardness is resistance to indentation. Toughness is how much energy the material absorbs before it fractures. A material can be hard and brittle, or soft and very tough, so choosing the wrong one of these as 'strong' is how a part that passed every specification arrives cracked. A part is only as good as the property nobody inspected.

TAKEAWAY 11W

Strength, stiffness, hardness, and toughness should not be treated as synonyms.

9.2 Why the acceptance limit is conservative

ORDNANCE & ENGINEERING · A PERSON · CHOICE

SCENE 34W

An inert support component has passed a controlled proof test, but the field procedure uses a lower operating limit. The crew wants to know why the margin is larger than the measurement uncertainty alone.

GUIDE 35W

Ask what changes outside the proof rig: temperature, alignment, repeated use, measurement error and handling variability. A written limit protects against combinations of ordinary deviations rather than assuming every input remains at its best-tested value.

ASK 25W

Why is the written operating limit — an administrative limit, set by people rather than by the metal — kept well below the proof-test boundary?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

real operating conditions vary from a controlled qualification test

VERDICT 63W

A proof test establishes what happened in one controlled configuration. Real use adds temperature changes, alignment error, wear, measurement uncertainty and ordinary handling variation. A written operating limit sits inside the demonstrated boundary so one credible deviation does not erase the full margin. The general engineering lesson is that a safety factor belongs to the system of uncertainties, not to one number alone.

TAKEAWAY 14W

Conservative limits create margin against combinations of ordinary uncertainty, handling variation, and environmental change.

9.3 High-speed imaging and radiography

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A ROOM · BALLPARK

SCENE 41W

On the analysis table are high-speed photographs from yesterday's implosion test, with radiographs of the opaque assembly beside them. Each image is labeled with its exposure time. Bacher has listed the object speeds and the feature sizes his committee wants measured.

GUIDE 47W

Four numbers, and two of them are exposure times a thousand apart. Ask of each which one this camera used. And note what the answer is: a limit the optics cannot beat. Anything moving during the exposure travels a real distance, and that distance is the blur.

ASK 4W

Estimate the blur distance.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a photograph records everything that happened while the shutter was open

VERDICT 80W

Anything moving during the exposure travels a real distance, and that distance becomes motion blur: speed times exposure time. 500 metres per second across 1 microsecond is half a millimetre. That is a resolution limit even if the optics are perfect. Exposure time, viewing angle, penetration, and timing all decide what an image can establish. Blur is especially useful because it is calculable before the shot, so the diagnostic can be designed around the feature size rather than judged afterwards.

TAKEAWAY 11W

Exposure time sets a direct spatial-blur limit for rapidly moving objects.

9.4 The nucleus as a physical system — Review

THEORY & CALCULATIONS · A ROOM · CALLBACK · CHOICE

SCENE 48W

A crate reaches the stockroom with a label on each end, and the two do not describe the same nucleus. One counts 92 protons and 146 neutrons. The other reads uranium 235. Hans Bethe, who heads the Theoretical Division, wants it settled before the material is booked in.

GUIDE 55W

One label counts particles and the other is a nuclide name, so put them in the same currency before comparing. A mass number is protons and neutrons added together. The element is fixed by the proton count alone, and that part is not in dispute here. What is in dispute is what the nucleus weighs.

ASK 18W

The two labels on the crate give different counts for the same nucleus. What does the disagreement mean?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

an atom has a nucleus of protons and neutrons, with electrons around it

VERDICT 83W

A nuclide symbol carries two facts and they are not interchangeable. The element name is the proton count, and that alone settles the chemistry. The mass number is protons and neutrons together. So 92 protons with 146 neutrons is mass 238, and the crate is uranium 238 by one label and uranium 235 by the other. Same element, same behaviour in a beaker, a different nucleus and a very different cross section. No chemical assay on this mesa would tell the two apart.

TAKEAWAY 18W

> Chemical identity and nuclear identity are separate facts, and a label stating only one of them is incomplete.

END OF THE STAGE 21W

A surrogate experiment is powerful only when the claim it is allowed to support is written down before the result arrives.

STAGE 10 OF 15

Everything Works Alone

PLAN CARD 103W

It is February 1945, and the joined system has failed even though each part passed alone. The fault may sit at an interface, across a batch, or inside the tools that measure time. The British Mission on the Hill has been asked for a check that nobody here made. Two batches of the same part went in, and nobody wrote down which was which. Every group can show that its own piece was good, and that is the problem. Today you must decide which shared claim broke when the parts met. Until you know, another test may fail for the same hidden reason.

BRIEFING 41W

Bench tests pass, but the integrated system does not. The player compares physical timescales in an inert test, chooses population-level qualification evidence instead of a champion prototype, and combines independent diagnostic timing errors without turning component synchronization into a weapon recipe.

WHAT YOU DECIDE 21W

Diagnose a system that fails only when its individually qualified parts are connected, using timing, population testing, and independent error budgets.

10.1 How far can a timing mismatch propagate?

THEORY & CALCULATIONS · A ROOM · BALLPARK

SCENE 39W

An inert materials test sends a stress wave through a plate while two diagnostic channels start a fraction of a microsecond apart. The team wants to know whether that timing mismatch is large enough to matter to the measurement.

GUIDE 45W

Four numbers, and two of them are channel spreads a hundred apart. Ask of each which spread the measurement gave. And note why this matters at all. Several clocks run at once here, and a timing disagreement becomes a distance the wave has already travelled.

ASK 17W

Estimate the separation produced when two diagnostic triggers begin an inert stress-wave measurement at slightly different times.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a wave crosses a distance in a time set by its speed

VERDICT 59W

Distance equals speed times time. In the inert plate, a stress wave moving at 5×10^3 m/s travels 2.5 mm during a 0.5 microsecond timing mismatch. That converts a timing specification into the spatial error it creates in the measurement. The lesson applies to any fast diagnostic system: time synchronization matters because moving phenomena turn clock error into position error.

TAKEAWAY 15W

A small timing error becomes a spatial error when a physical front is moving quickly.

10.2 What evidence qualifies a production lot?

ORDNANCE & ENGINEERING · A PERSON · VALUE

SCENE 29W

A production lot of timing modules is waiting on a release decision. The bench holds one carefully built prototype, dimensional inspection records, and access to a large integrated trial.

GUIDE 39W

Ask what each option tells you about the units in the lot. A champion prototype measures design potential. Inspection measures conformance. A large integrated trial entangles many causes. The release claim is about the distribution of the manufactured population.

ASK 10W

Which measurement tells you whether a production lot is reliable?

BEHIND THE BUTTON 140W

Why a prototype cannot qualify a lot. It was built by the best person available, slowly, with attention nobody can repeat a thousand times. It measures what the design can do, not what the production line did. Those are different questions and the release decision is about the second.

What inspection records can and cannot show. They record that units passed the checks somebody chose. If the failure mode is not among those checks, a perfect record is consistent with a bad lot — which is why sampled functional testing exists at all.

Why the full-system trial is seductive and expensive. It exercises everything at once, so a pass is reassuring and a failure is hard to attribute. Spending most of the credit on one integrated result can leave you knowing that something worked without knowing whether this lot will.

TAKES AS READ

a sample can estimate the spread of a production population · a component test is more diagnostic than an integrated test when the component itself is the question

VERDICT 57W

A single excellent unit can be repeatable and still say nothing about the next units off the line. Lot acceptance is a population claim. Dimensional inspection checks conformance to a drawing, not functional timing. A large integrated trial mixes too many causes. Sampling the manufactured population exposes unit-to-unit and lot-to-lot spread and directly answers the release decision.

TAKEAWAY 18W

Reliability is a population property, so lot acceptance must sample the manufactured distribution rather than its best member.

10.3 Close the diagnostic timing budget

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A ROOM · BALLPARK

SCENE 24W

Three diagnostic timing contributions are on the board. Each subsystem lead calls theirs small, but the reconstruction cannot be better than the full chain.

GUIDE 47W

Four numbers, and one of them is the other three added straight. Ask whether independent errors add that way. They do not. They combine in quadrature, so the largest term dominates. The straight sum is offered because it is the intuitive answer, and it overstates the total.

ASK 12W

Estimate the combined uncertainty of independent timing contributions in the diagnostic chain.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

independent errors do not simply add

VERDICT 48W

Independent timing uncertainties combine in quadrature. Two, three and four nanoseconds therefore give about 5.4 ns rather than 9 ns. The largest term contributes the most variance and is the best place to improve first. This is metrology: the numbers describe the measurement system, not a firing sequence.

TAKEAWAY 16W

A diagnostic timing budget must include the complete measurement chain and distinguish independent from shared errors.

END OF THE STAGE 23W

Integration failures often live in interfaces and population variation, so qualification must test the connected system and the manufactured distribution—not just ideal components.

More Data, Less Confidence

PLAN CARD 98W

It is March 1945, and more data have made the lab less sure, not more. New tests exposed doubts that the old error bars hid. The mathematicians say those error bars were never added up properly in the first place. More tests were meant to settle this, and so far they have not. Some of the tools meant to check each other share a part, and can fail together. Today you must find which doubt controls the choice and which tools can fail as one. A weak pass now could freeze a bad part into the final system.

BRIEFING 38W

New measurements have narrowed some uncertainties and exposed others. The player combines independent terms correctly, follows conservation laws through fission fragments, diagnoses competing sources in an integration model, and applies a pre-declared rule to a borderline material measurement.

WHAT YOU DECIDE 21W

Build an uncertainty budget that ranks what limits the program, distinguishes independent from shared errors, and makes borderline acceptance decisions explicit.

11.1 Carrying uncertainty through

THEORY & CALCULATIONS · A ROOM · BALLPARK

SCENE 36W

A predicted reaction rate rests on a cross section known to a few per cent and a flux known to a few more. Bethe is asking the theory group how well that prediction is actually known.

GUIDE 46W

Four numbers, and one of them is the two others added straight. Ask whether independent uncertainties add that way. They do not: they combine in quadrature. And note what the answer is for. The larger term dominates, so improving the small one barely moves the total.

ASK 5W

Estimate the total fractional uncertainty.

BEHIND THE BUTTON 106W

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Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

two independent errors are unlikely to be wrong the same way at once

VERDICT 78W

Independent uncertainties combine in quadrature: square each contribution, add the squares, and take the root. 8% and 6% make 10, not 14. The larger term dominates, so reducing a small contribution may barely change the total while improving the largest one matters much more. That ranking is the practical value of an uncertainty budget. It tells a laboratory with limited people and time which measurement is worth improving first, rather than treating every error bar as equally urgent.

TAKEAWAY 14W

Adding independent uncertainties linearly is usually too pessimistic; ignoring correlation can be too optimistic.

11.2 Where the fragments go

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A PERSON · CHOICE

SCENE 45W

A nucleus at rest splits into a heavy fragment and a light one. The masses are measured and the bay is arguing about which fragment carries the larger share of kinetic energy. Bethe has asked for an answer before anyone looks at the detector data.

GUIDE 50W

Four answers, and the nucleus started at rest. Ask of each whether it respects that. Total momentum was zero, so the two fragments leave with equal and opposite momentum. From there the question is what equal momentum does to two different masses, and one option refuses to decide at all.

ASK 17W

A nucleus at rest splits into a heavy and a light fragment. Which carries more kinetic energy?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

momentum is conserved in an isolated system

VERDICT 72W

The nucleus starts at rest, so total momentum is zero. The two fragments therefore leave with equal and opposite momentum. The lighter fragment must move faster because the momentum magnitude is the same for both. Kinetic energy can be written as $p^2/(2m)$. At equal momentum, the smaller mass has the larger kinetic energy. Momentum conservation fixes the relation between the motions; the energy formula then tells how the released energy is divided.

TAKEAWAY 13W

Momentum conservation fixes the ratio of fragment speeds, and energy follows from it.

11.3 Build the integration uncertainty budget

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A ROOM · DIAGNOSIS

SCENE 28W

An inert integration trial has returned two anomalies, and the next cycle can afford one major push. Name the wrong source and weeks disappear into the wrong fix.

GUIDE 35W

The panel has more than one thing wrong, so no single cause covers it. Ask which candidate explains each reading and which leaves evidence unexplained. The budget's value is the ranking rather than one total.

ASK 20W

Read the whole reading panel, then select the two causes that together account for every reading. No single cause does.

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

two problems can have two different causes

VERDICT 48W

An uncertainty budget lists separate contributions from material properties, geometry, timing, diagnostics and models, then asks how strongly each moves the final claim. Two anomalies in different evidence chains need not share one cause. Forcing one story onto both can send the program to improve the wrong subsystem.

TAKEAWAY 12W

Improvement starts by naming and ranking the dominant independent sources of uncertainty.

11.4 Acceptance criteria

CHEMISTRY & METALLURGY · A ROOM · BALLPARK

SCENE 38W

Seaborg has a property that has to be at least 100 units and a measurement reading 102, plus or minus 3. A batch waits on the answer. Material this scarce cannot be scrapped casually, or accepted on hope.

GUIDE 50W

Four numbers, and two of them are uncertainties ten times apart. Ask of each which one this measurement carries. And note what gets compared with the threshold. Not the central value, but the lowest the result could plausibly be. That rule has to be agreed before the number is known.

ASK 8W

Estimate the conservative lower bound of the measurement.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a measurement has a range, not just a value

VERDICT 77W

An acceptance criterion turns a desired property into a rule: a limit, a measurement method, and a stated uncertainty. Results near the boundary are why the uncertainty rule must be agreed before the number is known. Here the central value is 102, but the lower plausible value is 99. That falls below the 100-unit requirement, so the measurement does not demonstrate a clean pass. Reading only the midpoint treats an uncertain measurement as though it were exact.

TAKEAWAY 10W

Acceptance criteria need explicit rules for uncertainty and borderline results.

END OF THE STAGE 17W

An uncertainty budget is a map of what to improve next, not decoration around a central estimate.

One Integrated Trial

PLAN CARD 98W

It is April 1945, and one hidden bottleneck now controls the whole schedule. Only a few full tests remain, and a dramatic test may not tell the two main ideas apart. The Army wants a date it can hold people to, and it wants it now. The work that sets that date is not the work most people are busy with. Today you must choose a useful test, find the work that sets the date, and limit what a small mockup can prove. The wrong choice could spend the last test and still leave the key question open.

BRIEFING 37W

The full program is converging on one integrated non-nuclear campaign. The player spends scarce test access on the experiment that can actually separate hypotheses, calculates schedule float, and states scaling limits before interpreting an inert hydrodynamic mockup.

WHAT YOU DECIDE 23W

Choose tests that can discriminate between explanations, expose the schedule's true critical path, and define what scaled inert experiments can and cannot transfer.

12.1 Buy the test that can decide

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A ROOM · VALUE

SCENE 40W

Three team leads are waiting outside the facility office, each with a test they say cannot wait. There is one week of access left in the schedule and only one proposal can take it. Bacher needs a name before Monday.

GUIDE 52W

One week of access, three tests, and one name needed before Monday. Open each proposal and ask what decision its result would change. A test that everybody agrees is important but whose result changes nothing this month has not earned the week. Commit to the one that moves the programme's next decision.

ASK 14W

Which test changes the programme's next decision, rather than only improving its counting statistics?

BEHIND THE BUTTON 108W

What makes a test decisive rather than valuable. Two possible outcomes leading to different actions. A test whose likely result confirms the current plan is worth running eventually and not worth the only week of access left.

Why the leads all say theirs cannot wait. Each is right about their own programme and none of them is holding the schedule. The scheduler is the only person looking at what the next decision actually requires, which is why the choice sits there.

Why Monday. A name given late costs the week itself, so an imperfect choice made in time beats a better one made after the access has lapsed.

TAKES AS READ

an experiment is useful when different explanations predict different outcomes · facility time is limited

VERDICT 75W

Before a test is run, each hypothesis should predict what the chosen observable will do. If both hypotheses predict the same trace, even perfect data cannot separate them. Repeating one condition can reduce statistical uncertainty while leaving the decision unchanged. A large integrated shot can also be ambiguous if several causes move the same output. The valuable test is the smallest one whose outcomes differ between the hypotheses, with diagnostics aimed directly at that difference.

TAKEAWAY 18W

A test is valuable when its possible outcomes distinguish competing explanations, not when it merely produces more data.

12.2 The chain that sets the date

ORDNANCE & ENGINEERING · A PERSON · BALLPARK

SCENE 48W

Groves, the Army officer who commands the Manhattan Project, wants a date for the integrated trial, and three chains of work are running in parallel toward it. Lens inspection, electrical qualification and casing work each report their own weeks, and nobody has put the three on one sheet.

GUIDE 52W

Four numbers, and two of them belong to other questions: one task, and both chains added together. Ask of each whether it is a chain length. Only the longest chain moves the date. The gap between the two is what a parallel chain can lose before it becomes the one that matters.

ASK 13W

Estimate how much a parallel chain can slip before it controls the date.

BEHIND THE BUTTON 106W

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Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

work that can happen in parallel does not add up

VERDICT 88W

The longest chain is the critical path. It is the only place where adding people or shifts moves the date. Effort poured into a task that is not on it changes nothing, however busy it makes everyone. It is spent on float that was already there. Float is the gap between the controlling chain and the one being worked on, and it is finite. Enough slippage on a parallel chain makes that chain the new critical path. At which point the whole calculation has to be done again.

TAKEAWAY 8W

Schedule management starts with dependencies, not activity counts.

12.3 What a scaled inert experiment can and cannot tell you

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A ROOM · CHOICE

SCENE 34W

Bacher has a half-scale inert mockup on the table. Geometric dimensions, propagation measurements and diagnostic records are listed at both scales. He wants the team to state the similarity assumptions before interpreting the result.

GUIDE 29W

Ratios can transfer when the relevant dimensionless groups are matched. Intrinsic material lengths and times do not automatically scale with the geometry. Choose the claim that preserves that distinction.

ASK 16W

A half-scale inert hydrodynamic mockup is run. Which conclusions are entitled to transfer to full scale?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a dimensionless ratio has the same value at any scale · pressure, temperature and equations of state — taken as read

VERDICT 60W

Similarity is controlled by dimensionless groups. If the governing ratios match, a scaled inert experiment can reproduce the behavior associated with them. But material diffusion lengths, relaxation times and propagation properties have their own dimensions and do not simply halve because the geometry does. The scientific discipline is to write down which similarities are preserved before the result is used.

TAKEAWAY 23W

A smaller model is useful only when the lab states which parts of the physics stay alike and which need a fresh test.

12.4 What the shop is making today

THEORY & CALCULATIONS · A ROOM · SPOT

SCENE 41W

The machine shop takes work from every division and cannot take it all. The priority is set on a card at the window. Bacher's committee changes it whenever the schedule does, after a test date moves or an assay comes back.

GUIDE 51W

Release the jobs the current priority wants and hold the rest. The card at the window is changed during the day and nobody announces it. What is scored is the jobs either side of a change, because they are the only ones that show whether the window is reading the card.

ASK 12W

Release to the priority on the card, and keep watching the card.

BEHIND THE BUTTON 100W

Why a shop priority changes. Early it is whatever the test date needs. When an assay comes back wrong it is whatever the re-measurement needs, because a number nobody trusts stops every calculation resting on it. When a division head appears it is whatever that division wants, which is a different rule and worth naming honestly.

Why the changeover costs here. A machined part is hours of a scarce machinist and a scarcer piece of material. A job released under the old priority is not merely late — it has consumed the capacity the new priority was written to buy.

TAKES AS READ

a workshop with limited capacity decides what to make first

VERDICT 131W

Three priorities run across the day and each wants a different part of the queue. Anything the test date needs, then anything a re-measurement needs, then anything a division head has asked for in person. A job can answer two at once, which is what makes the change cost real. The panel scores the window either side of each change, because most of the queue is wanted by neither priority and is correctly held by somebody who has read nothing. And the cost here is material as well as time: a part released under yesterday's priority has used a machinist, a lathe and a piece of metal that the new priority was written to spend on something else, and none of the three comes back at the end of the day.

TAKEAWAY 15W

The cost of a withdrawn instruction is paid by whoever is still working to it.

END OF THE STAGE 29W

The best integrated test is the one whose possible outcomes change the decision, and the best schedule is the one that exposes the dependency that really sets the date.

Write the Questions Before Trinity

BEFORE THE STAGE — CANVASS

Who is worried about the same thing you are

Ask each division about the problem so you know whether anyone has already solved part of it.

PLAN CARD 96W

It is July 1945, and the Trinity test can happen only once. A missing tool or a vague question cannot be fixed after the event. The lab wants each question written down before the shot rather than after it. The site is a stretch of desert with ranches and small towns downwind of it. People and land outside the site may bear effects they did not choose. Today you must decide what proof each claim needs and what must be watched beyond the test field. A test that is asked nothing proves only that something happened.

BRIEFING 48W

A single full-scale test is about to carry years of expectations. The player must prevent the event from becoming its own proof by writing the questions first, establishing independent diagnostics and environmental measurements, and separating what an integrated outcome can show from what only post-event monitoring can reveal.

WHAT YOU DECIDE 20W

Treat Trinity as a one-shot scientific experiment by defining its questions, diagnostics, environmental monitoring, and model comparisons before the event.

13.1 Write Trinity as an experiment

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A ROOM · PROTOCOL

SCENE 32W

Bethe wants Trinity written down as questions before it becomes an event. One integrated test carries dozens of expectations, and a dramatic outcome cannot be allowed to count as proof of everything.

GUIDE 47W

Four questions and four kinds of evidence. Pair them by asking what each question needs: the whole event, one subsystem's traces, a comparison with something written beforehand, or measurements taken afterwards. A test without its questions written down first becomes an argument later about what it proved.

ASK 12W

Match each pre-test question to the evidence that can actually answer it.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a test proves what it was designed to ask

VERDICT 67W

Different claims require different evidence. The integrated event can answer whether the full system functioned at all. Independent diagnostic records show whether the measurement system itself remained interpretable. A model is tested only against a prediction written before the observation. Consequences beyond the instrument field require measurements after the event and outside the apparatus. Writing those questions first prevents one spectacular outcome from swallowing every other claim.

TAKEAWAY 22W

> A one-shot full-system test should be written as a set of questions before the event, including questions about environmental and human consequences.

13.2 Integrate a one-shot field experiment

ORDNANCE & ENGINEERING · A PERSON · SEQUENCE

SCENE 39W

The desert site has power, telephones, cameras, weather instruments and radiation monitors spread over many miles. Bad weather is already eating into the schedule. Bradbury, the Trinity assembly leader, wants every interface proved while it can still be fixed.

GUIDE 39W

Keep the sequence about experimental readiness. Establish site services and monitoring first, verify diagnostics and calibrations second, lock the approved test configuration only after those checks pass, and run only against explicit hold points with post-event monitoring already assigned.

ASK 20W

Arrange the field-experiment steps so that diagnostics, calibration records, safety and environmental monitoring are all verified before the irreversible event.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a schedule set by weather cannot be extended

VERDICT 67W

A one-shot field experiment is only useful if the measurement and safety systems are ready before the event. Site services, controlled access, weather and environmental monitoring come first. Diagnostic channels and calibrations are then checked end to end. The approved experimental configuration is locked only after those checks pass. Finally, the run proceeds against hold points, followed by the monitoring and documentation already assigned before the event.

TAKEAWAY 17W

Field integration should preserve the ability to detect and correct a failed interface before an irreversible test.

13.3 The plot that makes a decay a straight line

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A ROOM · CHOICE

SCENE 41W

Two counters have produced a table of count rate against time. The notebook has it plotted on ordinary axes, as a curve nobody can read a number off. Bethe wants the half-life out of it before the tower crew drives out.

GUIDE 57W

Four ways to plot the same data. Ask of each what shape it turns an exponential into. Then ask what the slope of that shape means. One option takes the logarithm of both axes, which suits a power law rather than a decay. And a straight line is easy to check by eye, which is the point.

ASK 19W

Which transformation makes exponential decay a straight line, and what physical quantity comes from the slope of the line?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a logarithm turns a constant ratio into a constant difference

VERDICT 76W

An exponential plotted on linear axes is a curve, which is hard to check by eye. Take the logarithm of the count rate and the same data becomes a straight line. Its slope is minus the decay constant, so the half-life follows from $t_{½} = \ln 2/\lambda$. The transformed plot also exposes failures: a second component can change the slope, while background can flatten the line at late times. Choosing the axes is part of the measurement.

TAKEAWAY 17W

> Plotting a logarithm turns an exponential into a straight line, and its slope is the decay constant.

13.4 Where the energy in fission comes from

THEORY & CALCULATIONS · A ROOM · CHOICE

SCENE 34W

A visitor at Bethe's blackboard asks why splitting a heavy nucleus releases energy while splitting a light one costs energy. The binding-energy curve is already drawn there, and nobody has used it to answer.

GUIDE 52W

Four accounts of where the energy comes from. Ask of each whether it uses the shape of the binding-energy curve. It rises steeply, peaks near iron, and falls slowly to uranium. So what matters is whether the products sit higher on it than the reactants, and two of these options ignore that.

ASK 12W

Why does splitting uranium release energy while splitting carbon would absorb it?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

a nucleus is held together by binding energy

VERDICT 87W

Binding energy per nucleon rises steeply from hydrogen, peaks around iron and nickel, and falls slowly to uranium. A reaction releases energy when its products sit HIGHER on that curve than its reactants, because the products are more tightly bound. Uranium splitting into two mid-mass fragments moves up the curve, so energy comes out. Carbon splitting would move down it, so energy would have to go in. The same curve is why fusion of light nuclei also releases energy — from the other side of the peak.

TAKEAWAY 11W

> The binding-energy-per-nucleon curve says which way a nuclear reaction releases energy.

END OF THE STAGE 19W

A one-shot experiment must define its questions and evidence in advance, including measurements of consequences outside the apparatus itself.

STAGE 14 OF 15

What Did Trinity Actually Prove?

PLAN CARD 95W

It is late July 1945, and Trinity worked, but the records do not all agree. Some tools match, one estimate stands apart, and fallout was found beyond the site. Observers forty miles off at Carrizozo were reading counters as the cloud went over. Cattle on ranches north of the site have burns along their backs. One test that works is not the same as a model that has been checked. Today you must decide which claims grew stronger and which still need a new check. Success cannot erase doubt, or harm outside the test field.

BRIEFING 48W

Trinity succeeded as an integrated test, but success is not the same as validating every model or eliminating every uncertainty. The player interprets a prediction-band comparison, distinguishes isotope physics from chemistry, spends follow-up work on independent checks, and controls configuration changes without turning post-test evidence into a recipe.

WHAT YOU DECIDE 21W

Compare observations with broad prediction bands, audit disagreements with independent evidence, and freeze only the claims that the test actually supports.

14.1 Compare a prediction band with an observation

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A ROOM · BALLPARK

SCENE 37W

Bethe has the theory division's prediction band chalked up beside a single diagnostic reading of 21 units, plus or minus 2. The temptation after Trinity is to treat one comparison as a verdict on the whole model.

GUIDE 48W

Four numbers, and two of them describe the band rather than the measurement: its midpoint and its lower edge. Ask of each what the comparison needs. A gap is measured in units of the measurement's own uncertainty. And note that a wide band would have accommodated almost anything.

ASK 15W

Estimate the offset in units of the measurement uncertainty, then state the strongest justified conclusion.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a wide prediction is easy to agree with

VERDICT 83W

The gap between an observation and a prediction is measured in units of the measurement's own uncertainty. Take the difference and divide by that uncertainty. Half a sigma from the middle of the band is consistent. But consistency with a band this wide is weak evidence, because 15 to 25 would have accommodated almost anything. Agreement supports the part of the model that produced the band, and no more. One agreeing diagnostic can sit beside others that disagree, and those still need explaining.

TAKEAWAY 13W

Evidence should update a model claim by claim, not by declaring total victory.

14.2 Why separation is a physical problem

CHEMISTRY & METALLURGY · A PERSON · CHOICE

SCENE 40W

At Chemistry and Metallurgy a production planner has brought Seaborg a request to enrich uranium-235 relative to uranium-238 by an ordinary chemical route. He has to decide whether the two isotopes give chemistry enough contrast to make that plan practical.

GUIDE 51W

Four explanations, and all four mention a difference. Ask of each whether that difference is chemical or physical. The electron structures are almost identical, which is what ordinary chemistry answers to. What is left is about one part in 235 of mass, and that is what a physical process can grip.

ASK 14W

Why is ordinary chemistry a poor way to separate uranium-235 from uranium-238 in bulk?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

chemical behaviour is set by electrons, and isotopes differ only in the nucleus

VERDICT 77W

Chemical behaviour is controlled mainly by electrons, and the two uranium isotopes have the same electron count and almost the same electronic structure. Their ordinary chemical reactions are therefore nearly identical. The useful contrast is the small mass difference, about 1 part in 235, which can be exploited by processes that respond to mass or mass-to-charge ratio. The tiny size of that contrast is why isotope enrichment requires repeated separation stages rather than one ordinary chemical reaction.

TAKEAWAY 20W

Isotopes of one element have nearly the same electron structure, so practical enrichment must exploit a tiny isotope-dependent physical difference.

14.3 Spend the follow-up checks on independence

THEORY & CALCULATIONS · A ROOM · VALUE

SCENE 47W

Fermi, a senior physicist known for quick estimates, has a blast estimate that disagrees with the theory prediction by three standard deviations. The division can complete two follow-up checks before the next review, and every extra run that repeats the same dependency consumes one of those chances.

GUIDE 50W

Two follow-up checks, and a three-sigma disagreement between theory and experiment. Open each check and ask what it depends on. A run that repeats a dependency the disagreement already contains cannot resolve it, however much data it returns. Buy the two that could change the interpretation rather than the precision.

ASK 10W

Which two checks can change the interpretation of the disagreement?

BEHIND THE BUTTON 129W

What three standard deviations is and is not. It is unlikely to be chance and it is not proof of new physics. Most disagreements that size turn out to be a shared assumption — a cross-section, a calibration, a background model — used on both sides or wrong on one.

Why repeating the measurement rarely helps. More runs on the same apparatus reduce the statistical part of the error and leave every systematic exactly where it was. If the disagreement is systematic, better statistics make it sharper and no more explicable.

What independence looks like here. A check that uses a different detector, a different reaction, or a different calibration path constrains the same quantity through different assumptions. Only that can tell a real discrepancy from a shared mistake.

TAKES AS READ

a systematic error does not average away with more repeats · independent measurements should not share the same critical calibration

VERDICT 77W

More of the same data reduces random error but carries the same systematic dependencies. Retuning the model can hide the discrepancy without explaining it. First audit the calibration shared by the measurements, because a common bias can create a confident disagreement. Then use a measurement based on different hardware or a different calibration path. If the discrepancy survives independent evidence, the case for missing physics strengthens. If it disappears, the original measurement chain becomes the leading suspect.

TAKEAWAY 19W

When theory and experiment disagree, the next measurements should attack shared systematics and add evidence based on different dependencies.

14.4 Freeze claims, not confidence

ORDNANCE & ENGINEERING · A ROOM · SEQUENCE

SCENE 29W

Trinity has produced new evidence and teams are already proposing last-minute changes. Bradbury wants the laboratory to separate what the data support from what people merely want to improve.

GUIDE 22W

Reconcile the records, compare them with pre-test predictions, name unresolved discrepancies, and only then approve evidence-supported changes and freeze the documented configuration.

ASK 14W

Order the post-test evidence process from raw records to a controlled final technical configuration.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

what was tested and what gets built can drift apart

VERDICT 58W

Separate records have to be reconciled before they can support one account of the event. That account is compared with predictions written beforehand, and unexplained discrepancies are kept visible rather than absorbed into a success narrative. Only then are evidence-supported changes approved and the technical documentation frozen. The lesson is configuration control and epistemic discipline, not weapon assembly.

TAKEAWAY 15W

Post-test configuration control should freeze only changes supported by reconciled evidence, while preserving unresolved discrepancies.

END OF THE STAGE 17W

A successful integrated test validates claims selectively; it does not erase discrepancies, environmental consequences, or model uncertainty.

STAGE 15 OF 15

What Can Science Decide?

PLAN CARD 93W

It is August 1945, and the lab's technical work is ending while the moral fight grows. Scientists can agree on the data and still disagree on what a nation should do. The lab has to hand on a record that later readers are able to test. Some of the people who built this want it shown once; some want it never used again. Today you must sign a record of what is known, what is in doubt, and who was harmed. Your last choice is where scientific proof ends and public policy begins.

BRIEFING 50W

By August 1945 the immediate technical program is closing, but scientists across the Manhattan Project disagree about demonstration, combat use, secrecy, and postwar control. The player finishes the evidence record, preserves diagnostic and materials capability, and signs a final statement that distinguishes what the laboratory knows from what it recommends.

WHAT YOU DECIDE 20W

Close the wartime evidence chain while separating technical findings, uncertainty, human consequences, and policy judgments that science alone cannot settle.

15.1 What the binding-energy curve can tell you

THEORY & CALCULATIONS · A ROOM · CHOICE

SCENE 24W

The binding-energy curve is still on Bethe's blackboard in August 1945. A younger physicist asks what the curve itself establishes—and what it does not.

GUIDE 36W

The curve compares how tightly bound nuclei are. Moving toward more tightly bound products can release energy. It does not, by itself, tell you whether a reaction will occur quickly or how to make it occur.

ASK 24W

Use the binding-energy-per-nucleon curve to decide which direction of nuclear change can release energy, without turning that principle into an ignition or weapon-design lesson.

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

like charges repel, and the nuclear force acts only at very short range

VERDICT 59W

Binding energy per nucleon rises from very light nuclei toward iron and nickel and then falls gradually for heavier nuclei. Moving toward more tightly bound products can release energy through a mass difference. But an energy-releasing direction is not the same thing as an accessible reaction path. The curve tells you the energy bookkeeping, not the kinetics or engineering.

TAKEAWAY 8W

Energetic favorability and reaction accessibility are different questions.

15.2 Final diagnostic readiness review

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A PERSON · SEQUENCE

SCENE 44W

Bacher has called the final readiness review, and the question in front of it is whether the instruments will actually record what happens. There is one event and no repeat shot. A channel that saturates or loses its timing erases an answer for good.

GUIDE 49W

All four steps happen, so ask what each one proves that the others cannot. An instrument nobody can describe afterwards produces data nobody can use. A chain of working parts is not a working chain. And independent coverage means one failure does not take a whole answer with it.

ASK 12W

Arrange the four cards from earliest cause or prerequisite to latest result.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

an instrument can be working and still miss the event

VERDICT 83W

Calibration and configuration records come first. An instrument nobody can describe afterwards produces data nobody can use. End-to-end timing and health checks then prove the chain works as a chain, rather than as a set of working parts. Independent coverage of the critical observables comes next — independent, so one failure does not take a whole answer with it. Deployment is approved with the residual risks named and a backup for each. Readiness means failures will be recognised, not that there are none.

TAKEAWAY 14W

The final product is a trustworthy evidence system, not merely a collection of sensors.

15.3 Leave a materials capability, not just better numbers

CHEMISTRY & METALLURGY · A ROOM · VALUE

SCENE 43W

Hornig's final dossier has to show more than that components existed. The division has one last hundred credits of improvement to spend, and the choice says what it thinks a materials capability is. It is the last money the division will be given.

GUIDE 51W

One hundred credits, and the choice says what the division thinks a materials capability is. Open each investment and ask what it leaves behind when the programme ends. Better numbers on the current components are worth something; a method, a standard or a trained group is worth it next year too.

ASK 8W

Which investment creates the most durable materials capability?

BEHIND THE BUTTON 120W

What a capability is, as distinct from a result. A result answers today's question. A capability answers questions nobody has asked yet: a calibrated technique, a documented procedure, a set of reference materials, people who know how to use them. The dossier is being read by people deciding what to fund next.

Why the temptation is to buy numbers. Precision on an existing measurement is visible, defensible and easy to write up. It is also spent the moment the component it describes is superseded.

Why this is the last hundred credits. Nothing follows it, so anything that needs a second round to be useful is worth nothing here. That constraint is what separates the durable investments from the promising ones.

TAKES AS READ

future batches may differ from today's batch · a record is useful when it preserves relationships, not only isolated measurements

VERDICT 78W

Another measurement can describe one batch more precisely, but it does not explain why the next batch differs. A process-to-properties record connects what was done to what the material became. That relationship can be tested across batches and used later to diagnose a change in structure or performance. Extra composition checks, tighter remeasurement, and higher throughput all have value, but none creates that causal record. The durable capability is the evidence chain linking process history to measured properties.

TAKEAWAY 15W

A durable materials programme links process conditions to composition, structure, properties, and traceable batch records.

15.4 Sign the final scientific record

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A ROOM · CHOICE

SCENE 39W

The last Evidence Chain is on the table. It contains the technical findings, their uncertainties, the observed consequences of Trinity, and a record of scientific disagreement about what should follow. Your signature decides what the laboratory claims as science.

GUIDE 31W

The final document has to do four things at once: state the technical findings, preserve uncertainty, include observed consequences beyond the laboratory, and label policy recommendations as recommendations rather than measurements.

ASK 6W

Which final statement can you sign?

BEHIND THE BUTTON 195W

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TAKES AS READ

technical findings, observed consequences, and policy judgments are related but not identical claims

VERDICT 52W

Scientists must say what the measurements show and how sure they are. They must also report harm beyond the test site, such as fallout and possible exposure. Scientists may argue for a policy, but that advice is not a measurement. The final record keeps facts, doubt, harm, and advice in separate parts.

TAKEAWAY 20W

The responsible scientific sign-off states evidence, uncertainty, and consequences clearly while marking the boundary between technical assessment and policy

> judgment.

END OF THE STAGE 19W

Scientific expertise carries a duty to state evidence, uncertainty, and consequences clearly while distinguishing those findings from political authority.

THE CLOSING CARD

The wartime Evidence Chain closes in August 1945, but the questions do not. Trinity worked on 16 July. People living near and downwind of the test had not been warned beforehand, and fallout was measured beyond the site. Hiroshima was bombed on 6 August and Nagasaki on 9 August. The Soviet Union entered the war against Japan on 8 August, and Japan announced its surrender on 15 August. Los Alamos did not become an empty mesa again; it became a permanent laboratory.

The technical record is real, and so are its limits. A successful test did not validate every model. Scientists across the Manhattan Project disagreed about demonstration, combat use, secrecy, and postwar control. Those were not disagreements that one more detector could settle. The people outside the fence who experienced displacement, secrecy, bombing, and fallout were part of the history even when they were not part of the experiment.

You closed the Evidence Chain by signing a record rather than a victory report. You kept measurement apart from inference, named the uncertainty instead of hiding it, preserved the evidence that forced the laboratory to change course, and stated the consequences that technical success does not erase. What you left later readers is a record they can test instead of a claim they are asked to trust.

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